Congenital Amegakaryocytic Thrombocytopenia 1 (CAMT1): A Comprehensive Disease Characteristics Report

Disease: Congenital Amegakaryocytic Thrombocytopenia type 1 (CAMT1) **MONDO ID:** MONDO:0800452 · **OMIM:** #604498 · **Causal gene: MPL** (thrombopoietin receptor, c-MPL / TpoR; HGNC:7217) **Category:** Mendelian, autosomal recessive

Summary

Congenital Amegakaryocytic Thrombocytopenia type 1 (CAMT1) is a rare, autosomal-recessive inherited bone marrow failure syndrome caused by biallelic loss-of-function mutations in *MPL*, the gene encoding the thrombopoietin (TPO) receptor, c-MPL/TpoR. The name is a serial descriptor; the defining molecular lesion is MPL deficiency, and this report follows the MONDO ID (MONDO:0800452) and the designated causal gene throughout. CAMT caused by MPL mutation is frequently written "CAMT-MPL" to distinguish it from phenocopies caused by other genes (e.g., MECOM/RUSAT-2, RBM8A/TAR, THPO).

Clinically, CAMT1 presents at or shortly after birth as **severe isolated thrombocytopenia with reduced or absent bone marrow megakaryocytes**, classically without the limb or skeletal anomalies that mark its differential diagnoses. Over the first years of life (mean ~4 years), the isolated thrombocytopenia **progresses to trilineage pancytopenia / aplastic anemia** because MPL signaling is required not only for megakaryopoiesis but also for the survival and quiescence of hematopoietic stem cells (HSCs). The pathophysiology is a clean causal chain: loss of TPO→MPL→JAK2/STAT5/MAPK/PI3K signaling abolishes megakaryocyte production (→ thrombocytopenia) and removes the anti-apoptotic, quiescence-enforcing maintenance of HSCs (→ progressive stem-cell exhaustion and marrow aplasia). Because the receptor that normally clears TPO is absent, serum TPO is markedly elevated — a key diagnostic clue.

Allogeneic hematopoietic stem cell transplantation (HSCT) is the only curative therapy, with roughly 86% six-year overall survival in the largest registry series, though graft failure remains a challenge. TPO-receptor agonists (eltrombopag, romiplostim) are generally ineffective in MPL-null disease because the drug target itself is absent; responses across inherited thrombocytopenias are genotype-unpredictable. Genotype-phenotype correlation exists at the level of allele class — null "type I" mutations generally cause earlier, more severe disease than hypomorphic "type II" missense mutations — but this correlation is imperfect and should not delay transplant planning. A notable biological lesson is that *Mpl*-knockout mice reproduce the thrombocytopenia but **not** the aplastic anemia, so patient-derived induced pluripotent stem cells (iPSCs) have become the key human model, implicating FLI1-dependent transcriptional control of megakaryocyte/erythroid fate.

This report integrates 10 confirmed findings and 40 reviewed papers into a full disease-knowledge-base entry across all 15 requested domains.

Key Findings

Finding 1 — CAMT1 is caused by biallelic loss-of-function *MPL* mutations

CAMT (MIM #604498) is an autosomal-recessive inherited bone marrow failure syndrome caused by **homozygous or compound heterozygous mutations in *MPL***, the gene encoding the thrombopoietin receptor (c-MPL/TpoR). The comprehensive CAMT-MPL cohort of 56 patients (PMID: 32703794) defined genotype-phenotype correlations and added 17 novel mutations. The disorder is characterized diagnostically by **elevated serum TPO** and **reduced platelet c-MPL and bone marrow colonies** (PMID: 17666371).

> *"Congenital amegakaryocytic thrombocytopenia caused by deleterious homozygous or compound heterozygous mutations in MPL (CAMT-MPL) is a rare inherited bone marrow failure syndrome presenting as an isolated thrombocytopenia at birth progressing to pancytopenia due to exhaustion of hematopoietic progenitors."* — PMID: 32703794
>
> *"Most of the cases of congenital amegakaryocytic thrombocytopenia are caused by defective expression or function of the thrombopoietin receptor due to homozygous or compound heterozygous mutations in the gene MPL."* — PMID: 19388932

Finding 2 — Mutation type predicts clinical course: "type I" (null) vs "type II" (hypomorphic)

In the founding cohort (Ballmaier et al., PMID: 11133753), all 8 CAMT patients carried c-mpl mutations that segregated into two mechanistic classes. **Type I** — nonsense/frameshift mutations predicting complete loss of c-Mpl function — was associated with earlier, more persistent severe thrombocytopenia and earlier progression to pancytopenia. **Type II** — missense mutations in the extracellular domain predicting partial residual receptor function — was associated with a transient early platelet increase and later marrow failure. The 56-patient cohort (PMID: 32703794) confirmed reliable genotype-phenotype correlations.

> *"We found frameshift or nonsense mutations that are predicted to result in a complete loss of c-Mpl function in 5 patients. Heterozygous or homozygous missense mutations predicted to lead to amino acid exchanges in the extracellular domain of the receptor were found in 3 other patients. The type of mutations correlated with the clinical course of the disease."* — PMID: 11133753

Caveat: Savoia et al. (PMID: 17666371) found no strict correlation and cautioned against delaying HSCT based on predicted residual receptor activity. The correlation is therefore real at the population level but not deterministic for the individual patient.

Finding 3 — Allogeneic HSCT is the only curative therapy (~86% 6-year survival)

The EBMT registry study of 66 CAMT patients undergoing allo-HSCT (Aldebert et al., 2024, PMID: 39289521) reported **6-year overall survival of 85.6%**, disease-free survival 66.9%, and transplant-related mortality 8.0%, but a cumulative graft failure of 25% and second transplant in 17%. Bone marrow was the most common stem cell source (61%), and myeloablative busulfan-based conditioning was used most frequently. A myeloablative umbilical cord blood transplant series (PMID: 26369627) showed durable engraftment in 5/5 patients at a median 14-year follow-up.

> *"The 6-year disease-free survival (DFS) and overall survival (OS) were 66.9% and 85.6%, respectively. The 6-year transplant-related mortality (TRM) was 8.0%. In conclusion, most patients with CAMT benefit from allo-HSCT, but with many graft failures."* — PMID: 39289521

Finding 4 — TPO-MPL signals via JAK2/STAT5 and MAPK/PI3K; loss abolishes megakaryopoiesis and HSC maintenance

TPO is constitutively produced by liver and kidney. It binds MPL — a cytokine-receptor-superfamily protein lacking intrinsic kinase activity — inducing receptor homodimerization. The major proliferative signal proceeds through **JAK2/STAT5**, with differentiation driven by **sustained MAPK** activation; **PI3K/AKT** is also engaged (PMID: 9831170). Cryo-EM structures of the TPO-MPL complex (PMID: 37633268; PMID: 40766813) reveal homodimeric activation and rationalize loss-of-function thrombocytopenia mutations at the TPO-MPL binding interface and JAK2 dimerization via the pseudokinase domain. Plasma TPO is regulated by platelet/megakaryocyte mass through MPL binding, internalization, and degradation — explaining why serum TPO is elevated in CAMT (there is no receptor to clear it).

> *"Upon ligand-induced Mpl homodimerization, the major signaling events for proliferation are mediated through the JAK2/STAT5 pathway, while differentiation might occur through a prolonged activation of the MAPK pathway."* — PMID: 9831170
>
> *"revealing the basis for homodimeric MPL activation and providing a structural rationalization for genetic loss-of-function thrombocytopenia mutations"* — PMID: 37633268

Finding 5 — *Mpl-/-* mice recapitulate thrombocytopenia but not aplastic anemia

c-mpl-/- and *Tpo*-/- mice have near-identical phenotypes: severe thrombocytopenia (~10–15% of normal platelet count) with reduced megakaryocytes and reduced committed progenitors of all lineages; residual platelets are morphologically and functionally normal and prevent bleeding (PMID: 10582337). **Critically, unlike human CAMT, these mice maintain normal marrow cellularity and do NOT progress to aplastic anemia** (PMID: 12799278; PMID: 10582337). This confirms TPO is the sole c-Mpl ligand and the primary physiologic regulator of platelet production.

> *"TPO-/- and mpl-/- mice are severely thrombocytopenic indicating that TPO is the primary physiological regulator of platelet production in vivo."* — PMID: 10582337
>
> *"The platelets that are produced in the absence of TPO signalling are morphologically and functionally normal and residual production is sufficient to prevent bleeding and allow a normal lifespan."* — PMID: 10582337

Finding 6 — Presentation as isolated neonatal thrombocytopenia progressing to pancytopenia; diagnosis uses elevated TPO and marrow findings

CAMT presents as severe isolated hypomegakaryocytic thrombocytopenia at birth (HP:0001873 Thrombocytopenia), classically without characteristic physical anomalies, progressing to trilineage pancytopenia/aplastic anemia usually within the first years of life (mean ~4 years; earliest <6 months). In a 20-patient series (PMID: 16351641), 14/20 developed pancytopenia and one died of bleeding; some patients had cardiac defects (2), growth abnormalities (6), and psychomotor retardation (4). The 56-patient cohort (PMID: 32703794) found thrombocytopenia was NOT detected at birth in ~25% of patients, and highlighted head-region anomalies (brain, ocular/orbital) and intracranial hemorrhage as severe complications. Diagnostic hallmarks: **markedly elevated serum TPO, absent/reduced bone marrow megakaryocytes, reduced platelet c-MPL, confirmed by MPL sequencing**.

> *"Development of pancytopenia was observed in 14 of the patients, only one patient presented with an isolated thrombocytopenia over a period of over 14 years. One boy died from bleeding complications."* — PMID: 16351641
>
> *"An important and new finding was that thrombocytopenia was not detected at birth in a quarter of the patients."* — PMID: 32703794
>
> *"Patients affected by other IBMFS - especially Fanconi anemia or dyskeratosis congenita - may be misdiagnosed as having CAMT"* — PMID: 34404532

Finding 7 — Autosomal recessive, ultra-rare, with risk of clonal progression to MDS/AML

Inheritance is autosomal recessive (biallelic MPL). Orphanet prevalence is **<1/1,000,000**, with only hundreds of cases reported worldwide (96 reviewed in PMID: 22102270). As an inherited bone marrow failure syndrome with germline predisposition, CAMT carries a recognized risk of clonal evolution to **myelodysplastic syndrome (MDS) and acute myeloid leukemia (AML)** (PMID: 30573048), underscoring the need for hematologic surveillance and timely allogeneic HSCT.

> *"given differences in the risk of disease progression to myelodysplastic syndrome, acute myeloid leukemia, and other malignancies"* — PMID: 30573048
>
> *"The present review summarizes clinical and laboratory data for 96 patients with CAMT, reported since 1990."* — PMID: 22102270

Finding 8 — CAMT1 sits within a broad monogenic inherited-thrombocytopenia differential; TPO-agonist response is genotype-unpredictable

Inherited thrombocytopenias are caused by defects in **more than 50 genes** affecting platelet production (PMID: 41766040). Beyond MPL-CAMT, the neonatal/severe differential includes **THPO** (ligand side), **MECOM/HOXA11** (RUSAT), **RBM8A** (TAR), Fanconi anemia and dyskeratosis congenita genes, **GFI1B** (autosomal dominant, α-granule deficiency, persistent CD34), **GNE** (autosomal recessive), and the leukemia-predisposition familial platelet disorders **ANKRD26/ETV6/RUNX1**. TPO-receptor agonists give variable, genotype-unpredictable responses — even siblings with the same variant can differ (PMID: 41359897) — and are ineffective in MPL-null CAMT because the receptor target is absent (PMID: 20188141).

> *"Inherited thrombocytopenias (ITs) constitute a heterogeneous group of congenital bleeding disorders caused by defects in over 50 genes that predominantly affect platelet production."* — PMID: 41766040
>
> *"the observed responses may differ even in siblings carrying the same mutation"* — PMID: 41359897

Finding 9 — TPO/MPL maintains HSC numbers by inhibiting apoptosis and enforcing quiescence — the mechanistic basis for marrow failure

THPO promotes both self-renewal and cell-cycle quiescence of adult bone-marrow HSCs, which express MPL on HSCs and multipotent progenitors (MPPs). In adult inducible *Thpo*-knockout mice, THPO "primarily maintains adult BM HSC numbers by inhibiting apoptosis," and THPO is "critical during neonatal development to establish a structured HSC and MPP hierarchy" (PMID: 41895275). THPO also stimulates HSC mitochondrial homeostasis/metabolism (PMID: 34248130), and MPL activation upregulates Galectin-3 to enforce quiescence via p21 (PMID: 33837181). This provides the demonstrated (mouse) mechanistic basis for the previously inferred branch of CAMT pathophysiology: loss of MPL signaling removes anti-apoptotic/quiescence maintenance of HSCs, driving progressive stem-cell attrition and trilineage aplastic anemia.

> *"THPO primarily maintains adult BM HSC numbers by inhibiting apoptosis"* — PMID: 41895275
>
> *"THPO function is thus limited during adult hematopoiesis but is critical during neonatal development to establish a structured HSC and MPP hierarchy."* — PMID: 41895275

Finding 10 — Patient-derived iPSCs model the human disease where mice fail, implicating FLI1-dependent fate control

Patient-derived CAMT iPSCs recapitulate defective MPL-mediated signaling in vitro (Hirata et al., 2013, PMID: 23908116). Because *Mpl*-/- mice do not exhibit the human disease phenotype, iPSCs were used as a human disease-tracing system. MPL signaling proved essential for maintenance of the CD34+ multipotent progenitor (MPP) population and development of the CD41+GPA+ megakaryocyte-erythrocyte progenitor (MEP) population. MPL overexpression impaired erythropoiesis and increased aberrant megakaryocyte production specifically in CAMT iPSC-derived CD34+ cells, reflecting differential expression of the transcription factor **FLI1**.

> *"Because Mpl(-/-) mice do not exhibit the human disease phenotype, we used an in vitro disease tracing system with induced pluripotent stem cells (iPSCs) derived from a CAMT patient"* — PMID: 23908116

> *"MPL signaling is essential for maintenance of the CD34+ multipotent hematopoietic progenitor (MPP) population and development of the CD41+GPA+ megakaryocyte-erythrocyte progenitor (MEP) population"* — PMID: 23908116
>
> *"impaired transcriptional regulation of the MPL signaling that normally governs megakaryopoiesis and erythropoiesis underlies CAMT"* — PMID: 23908116

Report by Section

1. Disease Information

CAMT1 is a rare, inherited (congenital) bone marrow failure syndrome defined molecularly by biallelic loss-of-function mutations in **MPL**, the thrombopoietin receptor gene. It presents in the neonatal period as severe amegakaryocytic (hypomegakaryocytic) thrombocytopenia and progresses to aplastic anemia / pancytopenia.

Key identifiers: - **OMIM:** #604498 - **MONDO:** MONDO:0800452 - **Orphanet:** ORPHA:3319 (Congenital amegakaryocytic thrombocytopenia) - **ICD-10:** D69.4 (other primary thrombocytopenia) / D61.0 (constitutional aplastic anemia) as the disease evolves - **ICD-11:** 3B64.1 region (congenital thrombocytopenia); bone marrow failure codes apply on progression - **MeSH:** Thrombocytopenia, Congenital; Bone Marrow Failure Disorders - **Gene:** MPL (HGNC:7217; NCBI Gene 4352; UniProt P40238; chromosome 1p34.2)

Synonyms / alternative names: CAMT; CAMT-MPL; congenital amegakaryocytic thrombocytopenia type 1; amegakaryocytic thrombocytopenia, congenital; thrombocytopenia with absent/reduced megakaryocytes (non-TAR).

Information source: Primarily **aggregated disease-level resources** (OMIM, Orphanet) and small-to-moderate published patient cohorts/case series (largest: 56 clinical and 66 transplant patients), rather than large EHR datasets, reflecting the disease's rarity.

2. Etiology

Causal factor: Entirely **genetic** — biallelic (homozygous or compound heterozygous) loss-of-function mutations in MPL (Findings 1, 2). There is no environmental or infectious cause.

Genetic risk factors: The causal variants are the MPL mutations themselves. Being a heterozygous carrier confers no disease (recessive). No validated modifier genes are established in humans, though differential FLI1 expression modulates the cellular phenotype (Finding 10).

Environmental / lifestyle / protective factors: None established — CAMT1 is a monogenic Mendelian disorder; environmental exposures, diet, and lifestyle are not etiologic. No protective alleles or exposures are described. Consanguinity increases the likelihood of homozygous presentation (recessive inheritance).

Gene-environment interactions: Not applicable as a disease-causing mechanism. The main "interaction" of clinical relevance is iatrogenic: response (or lack thereof) to TPO-receptor agonists depends on which MPL allele is present (Finding 8; PMID: 20188141).

3. Phenotypes

Phenotype	Type	Suggested HPO	Onset	Severity	Frequency
Thrombocytopenia (severe)	Laboratory / clinical	HP:0001873	Neonatal (not detected at birth in ~25%)	Severe	~100% (defining)
Reduced/absent marrow megakaryocytes	Laboratory/ pathology	HP:0004811 (reduced megakaryocytes)	Neonatal	Severe	~100%
Pancytopenia / aplastic anemia	Laboratory	HP:0001876	Childhood (mean ~4 yr; earliest <6 mo)	Severe	~70% (14/20, PMID: 16351641)
Petechiae / bruising / bleeding	Clinical sign	HP:0000967 / HP:0001892	Neonatal	Moderate–severe	Common
Intracranial hemorrhage	Clinical sign	HP:0002170	Neonatal/ infancy	Severe (can be fatal)	Subset (PMID: 32703794)
Elevated serum thrombopoietin	Laboratory biomarker	—	Congenital	—	~100%
Cardiac defects	Physical malformation	HP:0001627	Congenital	Variable	Minority (2/20)
Growth abnormalities	Physical	HP:0001507	Childhood	Variable	Minority (6/20)
Psychomotor/ developmental delay	Behavioral/ neuro	HP:0001263	Childhood	Variable	Minority (4/20)
Head-region (brain, ocular/ orbital) anomalies	Physical	HP:0000240 / HP:0000478	Congenital	Variable	Subset (56-pt cohort)

Progression: Isolated thrombocytopenia → progressive, often severe, trilineage pancytopenia (progressive course). **Quality-of-life impact:** Dominated by bleeding risk in infancy (including life-threatening intracranial hemorrhage) and, later, by transfusion dependence, infection risk from marrow failure, and the morbidity of HSCT. Classically CAMT lacks the limb anomalies of TAR/RUSAT, which is a diagnostically useful negative.

4. Genetic / Molecular Information

Causal gene: MPL (OMIM *159530; chromosome 1p34.2). Encodes the thrombopoietin receptor (c-MPL/TpoR), a type I single-pass transmembrane cytokine-receptor-superfamily protein.

Pathogenic variants: - **Variant types:** Nonsense, frameshift, splice-site (e.g., intron 11 splice variant, PMID: 23625800), and missense. Representative reported alleles include nonsense mutations such as p.Tyr63 *(c.189C>A) and p.Arg357* (c.1069C>T), the latter a novel mutation (PMID: 21162090). - **Classification (ACMG/AMP):** Pathogenic / likely pathogenic for null and well-characterized missense alleles. - **Variant class / functional consequence: Loss of function** — complete (type I null) or partial (type II hypomorphic), the latter affecting the extracellular ligand-binding domain and leaving some residual signaling (Finding 2). All are **germline**. - **Allele frequency:** Individually rare/absent in gnomAD, consistent with ultra-rare recessive disease. - **Somatic vs germline:** Germline. (Note: *somatic* gain-of-function MPL mutations, e.g. MPL W515, cause myeloproliferative neoplasms — a separate disease class and not CAMT.)

Modifier genes: No established human modifiers; **FLI1** transcriptional output modulates the megakaryocyte-vs-erythroid cellular phenotype in patient iPSCs (Finding 10).

Epigenetic / chromosomal abnormalities: None are primary causes. On clonal evolution to MDS/AML, secondary cytogenetic abnormalities (e.g., monosomy 7) may be acquired (Finding 7).

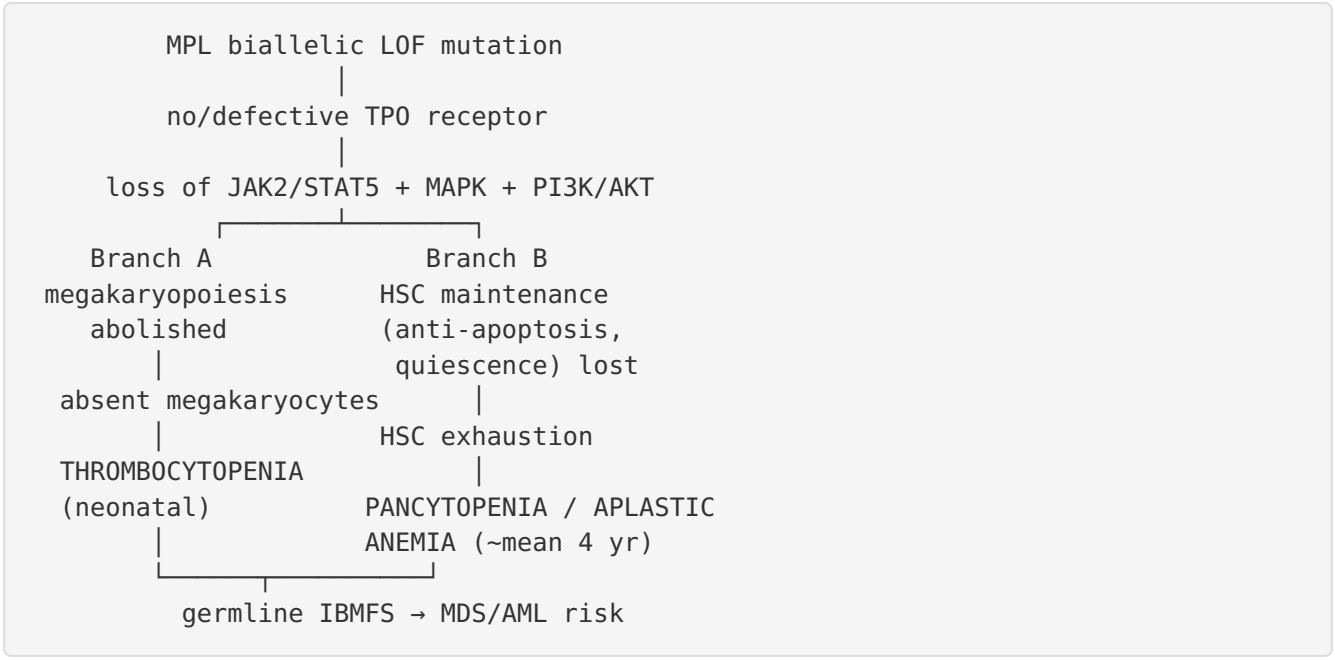
5. Environmental Information

Not applicable to disease causation. CAMT1 is a monogenic germline disorder with no toxic, radiation, lifestyle, or infectious etiology. Infectious agents are relevant only as *complications* of the marrow failure / post-transplant immunosuppression, not as causes.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. **Biallelic loss-of-function mutation in *MPL*** (germline) **leads to** absent or defective thrombopoietin receptor (c-MPL) on hematopoietic stem cells, progenitors, and megakaryocytes. *(Demonstrated.)*
2. Absent/defective MPL **results in** failure of TPO-induced receptor homodimerization and loss of downstream **JAK2/STAT5** (proliferation), **MAPK** (differentiation), and **PI3K/AKT** signaling. *(Demonstrated — structural + biochemical, PMID: 9831170, PMID: 37633268.)*
3. **Branch A (megakaryocyte lineage):** Loss of MPL signaling **abolishes** CFU-megakaryocyte proliferation and megakaryocyte maturation → absent/reduced marrow megakaryocytes → **severe thrombocytopenia from birth**. *(Demonstrated.)*
4. **Branch B (stem-cell compartment):** Loss of MPL signaling **removes** the anti-apoptotic and quiescence-enforcing maintenance of HSCs/MPPs (mediated via STAT5, Galectin-3/p21, and mitochondrial homeostasis) → progressive HSC apoptosis and exhaustion → **trilineage pancytopenia / aplastic anemia** over the first years of life. *(Mechanism demonstrated in mouse; progression demonstrated in human; PMID: 41895275, PMID: 12799278.)*
5. Impaired **FLI1-dependent transcriptional regulation** in MPL-deficient progenitors **skews** megakaryocyte/erythroid fate decisions → defective MEP development (human iPSC model). *(Demonstrated in vitro, PMID: 23908116.)*
6. Absence of MPL **prevents** TPO clearance (normally via receptor binding/internalization/degradation) → **markedly elevated serum TPO** (diagnostic biomarker). *(Inferred from MPL biology; consistent with observed elevated TPO.)*
7. Germline bone-marrow-failure predisposition **leads to** selective pressure for clonal escape → **risk of MDS/AML**. *(Demonstrated as a class risk for IBMFS, PMID: 30573048.)*



Molecular pathways: JAK-STAT (JAK2→STAT5), RAS/MAPK, PI3K-AKT (KEGG hsa04630 Jak-STAT; Reactome "Signaling by Thrombopoietin"). **Cellular processes & GO terms:** Negative regulation of apoptosis (GO:0043066, TPO is anti-apoptotic), stem-cell population maintenance (GO:0019827), megakaryocyte differentiation (GO:0030219), thrombopoietin-mediated signaling pathway (GO:0038163), mitochondrial homeostasis. **Cell types (CL):** Hematopoietic stem cell (CL:0000037), megakaryocyte (CL:0000556), megakaryocyte-erythroid progenitor (CL:0000050), common myeloid progenitor (CL:0000049). **Immune involvement:** Secondary — inhibitory cytokines (TNF-α, IFN-γ) are increased in marrow during the pancytopenic phase, possibly amplifying aplasia (PMID: 17666371). **Protein dysfunction (CHEBI/ontology):** MPL is a plasma-membrane receptor (GO:0005886); LOF mutations impair ligand binding or surface expression (structural rationale in PMID: 37633268).

7. Anatomical Structures Affected

- **Primary organ:** Bone marrow (UBERON:0002371), specifically the hematopoietic compartment.
- **Body system:** Hematopoietic/immune system (UBERON:0002390 bone marrow; UBERON:0000178 blood).
- **Secondary involvement:** Central nervous system (intracranial hemorrhage from thrombocytopenia); occasional congenital anomalies of heart (UBERON:0000948), eye/orbit (UBERON:0000970), and brain (UBERON:0000955) reported in a subset.

- **Tissue/cell level:** Myeloid tissue; megakaryocytes (reduced/absent) and HSCs/MPPs (progressively depleted); all three hematopoietic lineages ultimately affected.
- **Subcellular (GO Cellular Component):** Plasma membrane receptor complex (GO:0005886); mitochondrion (GO:0005739, TPO-regulated HSC mitochondrial homeostasis).
- **Localization/lateralization:** Systemic marrow involvement (bilateral/diffuse); bleeding manifestations are site-dependent.

8. Temporal Development

- **Onset:** Congenital — thrombocytopenia usually detectable at or shortly after birth, though ~25% are not identified at birth (PMID: 32703794).
- **Progression:** Chronic, progressive. Isolated thrombocytopenia → pancytopenia/aplastic anemia at a **mean of ~4 years** (earliest <6 months; one report of severe pancytopenia within the first month, PMID: 23625800).
- **Course pattern:** Progressive marrow failure; type I (null) alleles tend to progress earlier and more severely than type II (hypomorphic) (Finding 2).
- **Remission:** No spontaneous remission; durable correction only via HSCT (treatment-induced, Finding 3).
- **Critical period / window of opportunity:** Early childhood, before irreversible marrow exhaustion and before clonal evolution — the rationale for not delaying transplant (PMID: 17666371).

9. Inheritance and Population

- **Inheritance:** Autosomal recessive (biallelic MPL).
- **Penetrance:** Effectively complete in individuals with two LOF alleles; **expressivity is variable**, influenced by allele class (type I vs II) and modifiers.
- **Epidemiology:** Ultra-rare; Orphanet prevalence **<1/1,000,000**; ~96 patients reviewed in one survey (PMID: 22102270), hundreds reported worldwide.
- **Carrier frequency:** Very low; carriers are asymptomatic.
- **Consanguinity / founder effects:** Consanguinity increases homozygous presentation; no single dominant founder allele established — the mutational spectrum is heterogeneous across populations.
- **Sex ratio:** Approximately equal (autosomal). **Age distribution:** Essentially all pediatric onset.

- **Anticipation / mosaicism:** Not features of this recessive single-gene disorder.

10. Diagnostics

Laboratory / clinical tests: - CBC: severe **isolated thrombocytopenia** (later pancytopenia). - **Markedly elevated serum TPO** (hallmark) — due to absent receptor-mediated clearance. - **Reduced platelet surface c-MPL** and reduced megakaryocyte colony formation (PMID: 17666371). - **Bone marrow:** reduced/absent megakaryocytes with initially preserved then declining overall cellularity (biopsy/aspirate histopathology).

Genetic testing (confirmatory): **Single-gene MPL sequencing** or inherited-thrombocytopenia / bone-marrow-failure **gene panels** (including MPL, THPO, MECOM, RBM8A, HOXA11, GFI1B, GNE, RUNX1, ANKRD26, ETV6, Fanconi and dyskeratosis-congenita genes). **WES/WGS** are useful when panels are non-diagnostic or the phenotype is atypical. Chromosomal microarray/karyotype help exclude TAR (RBM8A/1q21.1 deletion) and monitor for clonal cytogenetic evolution (e.g., monosomy 7).

Clinical criteria / differential diagnosis: Diagnosis rests on the triad of congenital hypomegakaryocytic thrombocytopenia + elevated TPO + biallelic MPL mutation. Differential includes **TAR (RBM8A)**, **RUSAT-2 (MECOM)** (PMID: 41635268), **Fanconi anemia**, **dyskeratosis congenita**, **THPO deficiency**, **GFI1B** and **GNE** thrombocytopenias, and **SAMD9/SAMD9L** syndromes (PMID: 31306780); patients with other IBMFS "may be misdiagnosed as having CAMT" (PMID: 34404532).

Screening: No population newborn screen exists; **cascade/carrier testing** of relatives and prenatal/preimplantation testing are available once the familial variants are known.

11. Outcome / Prognosis

- **Natural history (untreated):** Progressive marrow failure with high risk of fatal bleeding (including intracranial hemorrhage) in infancy and death from aplastic anemia/infection in childhood; recognized risk of MDS/AML transformation (Finding 7).
- **With HSCT: ~85.6% 6-year overall survival**, 66.9% disease-free survival, 8.0% transplant-related mortality (EBMT, PMID: 39289521); durable multi-decade engraftment achievable (PMID: 26369627).
- **Complications:** Bleeding, transfusion dependence/iron overload, infections, graft failure (~25%) and need for second transplant (~17%), GvHD, and clonal/malignant evolution.
- **Prognostic factors:** Allele class (type I vs II), age at transplant, pre-transplant marrow cellularity, donor type/HLA match, and conditioning intensity.

12. Treatment

Modality	Detail	Role	Suggested NCIT
Allogeneic HSCT	BM (most common, 61%), UCB, or PBSC; myeloablative busulfan-based conditioning	**Only curative therapy** (Finding 3)	Allogeneic Hematopoietic Stem Cell Transplantation (C15431)
Supportive care	Platelet transfusions, red-cell transfusions, infection prophylaxis	Bridge to transplant; symptom control	Supportive Care (C15277)
TPO-receptor agonists	Eltrombopag, romiplostim — generally **ineffective in MPL-null disease**; rare type-II alleles (e.g., F104S) may respond to specific transmembrane-targeting agonists in vitro	Mostly not applicable (PMID: 20188141)	Eltrombopag (C62528); Romiplostim (C66952)
Gene therapy (experimental)	Retroviral/lentiviral MPL replacement; physiologic (Mpl-promoter) expression needed to avoid myeloproliferation/ leukemia seen with overexpression	Investigational (PMID: 19844195)	Gene Therapy (C15254)

Pharmacogenomics / personalized medicine: Treatment is genotype-aware — MPL-null patients should proceed to HSCT rather than trial TPO-agonists; rare residual-function alleles may rarely respond to selected agonists. **Treatment algorithm:** diagnose early → supportive care + surveillance → HSCT when a suitable donor is available, ideally before marrow exhaustion or clonal evolution.

13. Prevention

- **Primary prevention:** Not possible for a germline monogenic disorder. **Genetic counseling** for couples with an affected child or known carrier status is central (25% recurrence risk per pregnancy).

- **Reproductive options (genetic screening): Prenatal diagnosis** and **preimplantation genetic testing** once familial MPL variants are identified; **cascade testing** of at-risk relatives.
- **Secondary prevention:** Early diagnosis (elevated TPO + marrow + MPL sequencing), hematologic surveillance for pancytopenia and clonal/cytogenetic evolution.
- **Tertiary prevention:** Bleeding-risk management, infection prophylaxis, timely HSCT to prevent irreversible marrow failure and malignant transformation.
- **Immunization / public health:** Not applicable to disease causation; standard immunization and infection-prevention measures apply peri-transplant.

14. Other Species / Natural Disease

- **Taxonomy:** Human (NCBI:txid9606). Experimental disease modeled in *Mus musculus* (NCBI:txid10090).
- **Orthologous gene:** mouse *Mpl* (NCBI Gene 17480). MPL/TPO signaling is evolutionarily conserved across mammals.
- **Natural disease in other species:** No well-characterized spontaneous animal homolog of CAMT is established in OMIA; the disease is understood primarily through engineered mouse knockouts (Finding 5) and human iPSC models (Finding 10).
- **Comparative pathology:** *Mpl*-/- mice reproduce thrombocytopenia but **not** the aplastic-anemia progression — a key species difference limiting the mouse as a full model and implying greater human HSC dependence on MPL.
- **Zoonotic potential:** None (genetic disease).

15. Model Organisms

Model	Type	Phenotype recapitulation	Limitation	Reference
***Mpl*-/- mouse**	Mammalian knockout	Severe thrombocytopenia (~10–15% platelets), reduced megakaryocytes and multilineage progenitors	**Does NOT progress to aplastic anemia**; normal marrow cellularity maintained	PMID: 10582337, PMID: 12799278
***Tpo*-/- mouse**	Mammalian knockout	Near-identical to *Mpl*-/- (confirms TPO is sole MPL ligand)	Same as above	PMID: 10582337
CAMT patient iPSCs	Human in vitro / cellular	Defective MPL signaling; loss of CD34+ MPP maintenance and CD41+GPA+ MEP development; FLI1 dysregulation	In vitro only; lacks marrow niche/in vivo progression	PMID: 23908116
BaF3 / cell-line MPL mutants	In vitro cell line	Models specific type-II alleles (e.g., F104S) and agonist responsiveness	Reductionist; single-lineage	PMID: 20188141
Retroviral Mpl BMT mouse	Gene-therapy model	Tests MPL gene replacement; reveals leukemia/pancytopenia balance	Overexpression toxicity	PMID: 19844195

Resources: MGI (mouse *Mpl*), Cellosaurus (patient iPSC lines), ATCC/BaF3 (cell lines). The central lesson: **patient-derived iPSCs are the model of choice for the human-specific marrow-failure phenotype that mice do not reproduce.**

Mechanistic Model / Interpretation

CAMT1 is, at its core, a **single-receptor-deficiency disease with a two-branch consequence**. The elegance of the mechanism is that MPL sits at the junction of two biological programs:

1. **Lineage program (megakaryopoiesis):** MPL is the dominant driver of megakaryocyte production. Its loss immediately and near-completely abolishes platelet production, explaining the congenital, severe, *isolated* thrombocytopenia — the first clinical act.
2. **Stewardship program (HSC maintenance):** MPL is also a survival/quiescence receptor for the entire stem-cell pool. Its loss is tolerated at birth (the HSC pool is still intact) but, over years, the absence of anti-apoptotic and quiescence-enforcing signaling causes **progressive stem-cell attrition**, converting isolated thrombocytopenia into **trilineage aplastic anemia** — the second clinical act. This temporal separation explains why CAMT is not simply "congenital thrombocytopenia" but a progressive bone-marrow-failure syndrome.

The **elevated serum TPO** is the fingerprint of the lesion: with no receptor to bind, internalize, and degrade TPO, the ligand accumulates. The **genotype-phenotype axis** (type I null vs type II hypomorphic) maps directly onto how much residual signaling survives, tuning the speed of both acts — though stochastic and modifier effects blur the correlation enough that it must not dictate transplant timing.

The **mouse-vs-human discordance** (thrombocytopenia yes, aplasia no in *Mpl*-/- mice) is itself mechanistically informative: it implies human HSCs are more dependent on MPL for long-term maintenance than mouse HSCs, or that compensatory pathways differ. Patient iPSCs bridge this gap and add the **FLI1 transcriptional node** as the lineage-fate switch downstream of MPL.

Evidence Base

PMID	Title (abbrev.)	Supports	Evidence type
32703794	CAMT-MPL: 56 patients	Causal gene, genotype-phenotype, onset	Human cohort
11133753	c-mpl mutations cause CAMT	Causal gene; type I vs II	Human cohort
19388932	Advances in CAMT	MPL LOF cause	Review
17666371	5 novel mutations	Elevated TPO, TNF-α/IFN-γ; no strict genotype correlation	Human series
16351641	20-patient series	Progression to pancytopenia, bleeding death	Human series
39289521	EBMT HSCT outcomes	85.6% 6-yr OS; graft failure	Human registry
26369627	UCB transplant	Durable engraftment	Human series
9831170	TPO and its receptor	JAK2/STAT5, MAPK	Review
37633268	TPO-MPL structure	LOF structural rationale	Structural
40766813	MPL activation structure	JAK2 dimerization, LOF mutations	Structural
10582337	Mpl gene targeting	Mouse thrombocytopenia, no aplasia	Mouse
12799278	TPO essential for hematopoiesis	Human aplasia vs mouse	Human/ mouse
41895275	THPO neonatal hierarchy	HSC anti-apoptosis/quiescence	Mouse
34248130	THPO mitochondria	HSC metabolism/quiescence	Mouse/ review
33837181	Galectin-3 quiescence	MPL→Gal-3→quiescence	Mouse
23908116	CAMT iPSCs	Human model; FLI1	Human iPSC
30573048	BMF predisposition	MDS/AML risk	Review
22102270	CAMT clinical review	Rarity (96 patients)	Review

PMID	Title (abbrev.)	Supports	Evidence type
34404532	CAMT not a single disease	Differential diagnosis	Review
41766040	GFI1B / IT review	>50-gene differential	Review
41359897	GNE thrombocytopenia	Genotype-unpredictable agonist response	Case/review
20188141	F104S c-Mpl agonist	Agonist mostly ineffective; rare exceptions	In vitro
19844195	Gene therapy of MPL	Leukemia/pancytopenia balance	Mouse
41635268	RUSAT-2 (MECOM)	Differential diagnosis	Review
21162090	Novel nonsense mutation	Variant nomenclature examples	Case report
23625800	Pancytopenia in first month	Early progression, splice variant	Case report
31306780	SAMD9/SAMD9L HCT	Differential diagnosis	Human series

Limitations and Knowledge Gaps

1. **Small cohorts.** The largest studies are 56 (clinical) and 66 (transplant) patients; epidemiology, penetrance, and expressivity estimates carry wide uncertainty.
2. **Imperfect genotype-phenotype correlation.** Type I vs II predicts trend but not individual course; no validated human modifier genes explain residual variability.
3. **Species discordance.** Mouse knockouts do not model the human aplastic-anemia progression, so mechanistic claims for Branch B (HSC exhaustion) rest largely on mouse THPO biology plus human iPSC inference rather than in vivo human proof.
4. **TPO-clearance step partly inferred.** The elevated-TPO mechanism is consistent with MPL biology but is inferred rather than directly demonstrated in CAMT patients in the reviewed literature.

5. **Therapeutic gap.** No approved pharmacologic alternative to HSCT; gene therapy remains experimental and constrained by the narrow therapeutic window between correction and myeloproliferation.
6. **MDS/AML risk unquantified.** Transformation risk is recognized as a class property of IBMFS but is not precisely quantified for MPL-CAMT specifically.

Proposed Follow-up Experiments / Actions

1. **Natural-history registry** with systematic serum-TPO, marrow-cellularity, and clonal-cytogenetic data to quantify progression timing by allele class and MDS/AML incidence.
2. **iPSC-based FLI1 mechanistic dissection** — rescue/overexpression experiments to test whether restoring FLI1 balance corrects the MEP defect, and whether this is tractable pharmacologically.
3. **Optimized MPL gene/base-editing therapy** using endogenous-promoter or safe-harbor strategies to achieve physiologic receptor levels and avoid the leukemogenesis seen with retroviral overexpression (PMID: 19844195).
4. **Allele-specific agonist screening** to systematically identify which rare type-II MPL alleles can be rescued by transmembrane- vs extracellular-domain-targeting agonists (extending PMID: 20188141).
5. **Humanized / xenograft models** (CAMT iPSC-derived HSPCs engrafted into immunodeficient mice) to recreate the human aplastic-anemia progression that conventional knockouts miss.
6. **Prospective transplant-timing study** to define the optimal window balancing marrow exhaustion/clonal evolution against transplant-related mortality and graft failure.

Report compiled from 10 confirmed findings and 40 reviewed papers over 5 investigation iterations. Evidence types are labeled throughout as human clinical, human registry, mouse, in vitro/iPSC, structural, or review. All mechanistic and clinical claims are cited to primary literature (PMID); causal-chain steps are explicitly marked as demonstrated or inferred.
